

Sparse production radial-Gram decomposition, strengthened global owner

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-08-04 · this version issued 2026-08-04 · v1.1

1. Purpose and scope

R-166 v1.0 closed the R-165 amplitude annulus on one exact production principal fibre. Version 1.1 does not introduce a new theorem number. It extracts the stronger consequence already contained in the same rational certificate and records an exact boundary of the proof method.

On the fixed side-16 torus retain one strict-past antipodal pair at p , one at $2p$, and one fresh final pair at $4p$, with unit retained regulator multipliers and positive floor. Let

$$g = (g_p, g_{2p}) \in \mathbb{R}^{24}, \quad G = |g|, \quad \alpha \in \mathbb{R}^{12}. \quad (1.1)$$

For the exact R-153 endpoint-plus-sixth owner, excluding the explicit source Hessian,

$$\boxed{K_{\text{owner}}(g)[\alpha, \alpha] \geq \tilde{P}(G)|\alpha|^2}, \quad (1.2)$$

where

$$\boxed{\tilde{P}(G) = \frac{625}{21040201728}G^4 - \frac{25995}{87359488}G^2 - \frac{1323}{8192000}G - \frac{4402893}{30051663872}}. \quad (1.3)$$

The exact global certificate proves the stronger statement

$$\boxed{K_{\text{owner}}(g) > -\frac{4}{5}\mathbf{I} \quad \text{for every } g \in \mathbb{R}^{24}}. \quad (1.4)$$

Consequently the scoped R-164 reduction may take

$$\boxed{\rho = \frac{3}{20}}, \quad (1.5)$$

rather than the v1.0 choice $1/110$. This is still only one sparse $p:2p$ -past/fresh- $4p$ fibre. It does not close the complete multi-root owner, T-050, A13, a physical phase, a preferred morphology, a PDE, or Sector A.

2. Exact radial-Gram decomposition

Let P be the real orthogonal projector onto the complex doublet. For a six-real field w , set

$$d = |w|^2 + e, \quad \lambda = \frac{|Pw|^2}{d}, \quad b = 2Pw, \quad a = 2(Pw - \alpha_{\text{rad}}\lambda w), \quad \alpha_{\text{rad}} = \frac{5}{9}. \quad (2.1)$$

If B_J is the three-row Pauli-current Gram and B_L the conormal Gram, the R-130 Fierz identity gives

$$B_L = B_J + aa^T - bb^T. \quad (2.2)$$

The A1 coefficients are

$$c_0 = \frac{3}{250P_X}, \quad c_1 = \frac{243}{8000P_X}, \quad P_X = 4 + 10^{-12}, \quad (2.3)$$

and therefore

$$\boxed{B = (c_0 + c_1)B_J + c_1C, \quad C = aa^T - bb^T.} \quad (2.4)$$

This isolates the rational radial defect instead of applying an absolute six-row derivative envelope before using the favorable Pauli structure.

3. Exact Pauli resonance cancellation

Condition on the strict past and write

$$W = m + X, \quad n = \partial m, \quad z = S_{4p}\alpha, \quad u = \partial z. \quad (3.1)$$

For every real symmetric embedded Pauli generator S_A , put

$$J_A(t) = 2(W + tz)^T S_A(n + tu). \quad (3.2)$$

Then

$$J_A''(0) = 4z^T S_A u = 2\partial(z^T S_A z). \quad (3.3)$$

After the centered fresh expectation, $\mathbb{E}J_A(0) = 2m^T S_A n$ has frequencies of magnitude at most four, whereas (3.3) has only frequencies ± 8 and zero mean. Hence

$$\int \mathbb{E}[J_A(0)J_A''(0)] dx = 0. \quad (3.4)$$

The remaining current term is a square. The unaveraged Xn frequencies also do not resonate. The strict-past trace debit is retained in Section 6; (3.4) is not a positivity claim for the complete owner.

4. Floor-uniform radial correction

Put $R = (P - \lambda I)w$. Direct differentiation gives

$$\nabla \lambda = \frac{2R}{d}, \quad D^2 \lambda = \frac{2(P - \lambda I)}{d} - \frac{4(wR^T + Rw^T)}{d^2}. \quad (4.1)$$

Writing $x = |Pw|^2/d$ and $\delta = e/d$ yields

$$\frac{|R|^2}{d} = x(1 - x) - x^2\delta \leq \frac{1}{4}, \quad \frac{|w|^2}{d} \leq 1, \quad (4.2)$$

so

$$|w||\nabla \lambda| \leq 1, \quad |w|^2 \|D^2 \lambda\|_{\text{op}} \leq 6. \quad (4.3)$$

The removable origin is taken by continuity. From (2.1)–(4.3),

$$\|Da\| \leq \frac{38}{9}, \quad |D^2 a[z, z]| \leq \frac{80}{9} \frac{|z|^2}{|w|}. \quad (4.4)$$

For every real symmetric Q ,

$$|C : Q| \leq 8\|Q\|_{\text{op}}|w|^2, \quad (4.5)$$

$$|D(C : Q)[z]| \leq \frac{224}{9}\|Q\|_{\text{op}}|w||z|, \quad (4.6)$$

$$\left| \frac{1}{2}D^2(C : Q)[z, z] \right| \leq \frac{3208}{81}\|Q\|_{\text{op}}|z|^2. \quad (4.7)$$

Thus the paid outward constants are

$$L_C < \frac{189}{1000}, \quad H_C < \frac{1203}{4000}, \quad H_{\text{tr}} < \frac{3423}{8000}. \quad (4.8)$$

5. Direct single-pair harmonic coercivity

For the declared fresh $4p$ pair put $r = |m|^2$, $f = r^2$, and $q = |z|^2$. Then q has only modes $0, \pm 8$, $f_8 = r_4^2$, and positivity gives $q_0 \geq 2|q_8|$. Parseval gives

$$f_0 - |f_8| = r_0^2 + 2 \sum_{k=1}^3 |r_k|^2 + |r_4|^2 \geq r_0^2. \quad (5.1)$$

Hence

$$\boxed{\langle |m|^4 |z|^2 \rangle \geq \langle |m|^2 \rangle^2 \langle |z|^2 \rangle.} \quad (5.2)$$

The right side differs from the sharper but different decorrelation product $\langle |m|^4 \rangle \langle |z|^2 \rangle$ used in R-165. Gaussian fourth moments only add retained nonnegative terms, so the sixth Hessian contributes

$$A = \frac{9}{10} \frac{(100/63)^2 (20/91)}{4096^2} = \frac{625}{21040201728}. \quad (5.3)$$

6. One-use coefficient ledger

Use the exact R-165 covariance bounds

$$\frac{100}{63}\mathbf{I} \leq \Gamma_p, \Gamma_{2p} \leq \frac{100}{31}\mathbf{I}, \quad \frac{20}{91}\mathbf{I} \leq \Gamma_{4p} \leq \frac{10}{43}\mathbf{I}, \quad \text{Tr } \Gamma_{4p} \leq \frac{60}{43}. \quad (6.1)$$

Also $p^2 < 5/32$, the past derivative covariance is at most $625/507904$, and $\sqrt{2(100/31)(60/43)} < 301/100$. Paying each correction value, first derivative, half-Hessian, fresh covariance, and strict-past trace once gives

$$\begin{aligned} B_2 &= \frac{25995}{87359488}, & B_1 &= \frac{1323}{8192000}, \\ D_{\text{fresh}} &= \frac{729}{30294016}, & D_{\text{trace}} &= \frac{85575}{698875904}, \end{aligned} \quad (6.2)$$

and

$$D_0 = \frac{4402893}{30051663872}. \quad (6.3)$$

The raw coefficient behind B_2 is

$$256 + 32 \cdot \frac{224}{9} + 8 \cdot \frac{3208}{81}. \quad (6.4)$$

The Pauli square is not counted as a reserve and the source $9\mathbf{I}/10$ remains outside K_{owner} .

7. Exact global minimum and stronger budget

Exact arithmetic gives

$$\tilde{P}'(70) = -\frac{1360786403}{1277632512000} < 0, \quad \tilde{P}'(71) = \frac{97738714417}{876455903232000} > 0, \quad (7.1)$$

and

$$\tilde{P}''(70) = \frac{6865745}{5962285056} > 0. \quad (7.2)$$

The derivative has a unique nonnegative zero $70 < G_* < 71$. Convexity on that interval gives

$$\tilde{P}(G_*) \geq -\frac{332863942666997}{439505584128000}. \quad (7.3)$$

The v1.0 margin above $-9/10$ was

$$\frac{62691083048203}{439505584128000} > 0. \quad (7.4)$$

The stronger observation is

$$\frac{4}{5} + \tilde{P}(G_*) \geq \boxed{\frac{18740524635403}{439505584128000}} > 0. \quad (7.5)$$

Therefore the maximum rho justified by this lower certificate is

$$\rho_{\text{eff}} = \frac{10}{11} + \tilde{P}_{\text{lower}} = \frac{733552471943033}{4834561425408000}. \quad (7.6)$$

For the clean choice $\rho = 3/20$ the remaining exact margin is

$$\rho_{\text{eff}} - \frac{3}{20} = \frac{8368258131833}{4834561425408000} > 0. \quad (7.7)$$

The inherited R-164 downstream quantities become

$$-\frac{1}{110} + \rho = \frac{31}{220} > \frac{1}{10}, \quad \epsilon_v = \frac{5}{11} - \frac{\rho}{4} = \frac{367}{880}. \quad (7.8)$$

8. Exact direct-harmonic non-tensorization

Equation (5.2) is tied to one declared fresh pair. It does not tensorize with coefficient one when a second fresh harmonic is admitted. Take

$$m = \cos(2\theta), \quad z = a \sin(4\theta) + b \sin(8\theta). \quad (8.1)$$

Fourier multiplication, with normalized phase average, gives

$$\langle m^4 z^2 \rangle = \frac{5}{32} a^2 + \frac{1}{4} ab + \frac{3}{16} b^2, \quad (8.2)$$

whereas

$$\langle m^2 \rangle^2 \langle z^2 \rangle = \frac{1}{8} (a^2 + b^2). \quad (8.3)$$

The generalized Gram is

$$M = \begin{pmatrix} 5/4 & 1 \\ 1 & 3/2 \end{pmatrix}, \quad \lambda_{\min}(M) = \frac{11 - \sqrt{65}}{8}. \quad (8.4)$$

In the rational direction $(a, b) = (1, -1)$,

$$\langle m^4 z^2 \rangle = \frac{3}{32}, \quad \langle m^2 \rangle^2 \langle z^2 \rangle = \frac{1}{4}, \quad \frac{N}{D} = \frac{3}{8} < 1. \quad (8.5)$$

Thus coefficient-one direct harmonic coercivity fails on the enlarged $\{\pm 4, \pm 8\}$ support. Even if every old adverse coefficient is optimistically retained and only A is replaced by $(3/8)A$, the resulting polynomial satisfies

$$P_{3/8}(116) = -\frac{100799462911238297}{50250138451968000}, \quad (8.6)$$

and

$$P_{3/8}(116) + \frac{10}{11} = -\frac{606292707503941267}{552751522971648000} < 0. \quad (8.7)$$

This is an exact NON-TENSORIZATION result for a proof route, registered as NG-2026-08-04-A13-R166-DIRECT-HARMONIC-COERCIVITY-TENSORIZATION. It does not refute the original R-166 fibre and is not a counterexample to the actual complete production owner; new positive cross terms may change that owner.

9. Production-incidence audit and route decision

The next proof object cannot be obtained by merely adding abstract harmonic coordinates. R-156 fixes the shifted chart and the formal pullback-Hessian term, R-153 fixes the final-root current, trace and sextic identities, and R-149 proves that the actual cross synthesis $K = S_x S_v^*$ is indispensable. The registered authorities do not yet pin one complete $p \rightarrow 2p \rightarrow 4p$ production cylinder: the effective syntheses and square-root bases, raw, past and future covariances, output/common-heat/root-visit incidence, and the placement of complement, historical-low, forest and returned-mean terms are not all specified together.

Therefore the next rigorous step is to freeze those data in one canonical finite cylinder and differentiate its exact scalar once. Only then should the forward, legal reverse, balanced, low, current, trace, future, forest, projected-force and sextic blocks be assembled and bounded once. In parallel, T-054 and T-052 should continue truth-first candidate evaluation; no BCC morphology or particular PDE is privileged.

An ignored internal hostile scan of simultaneous fresh $4p + 8p$ directions found a stable small negative sampled Rayleigh value near $-1.93 \cdot 10^{-5}$, so the stronger unproved assertion $K_{\text{owner}} \geq 0$ is not a safe target. The sampled value remains about 0.80 above $-4/5$ and has no interval or rational enclosure. It is a regression falsifier, not theorem evidence and not a T-050 counterexample.

10. Devil's-advocate review

1. **Objection: v1.1 changes the polynomial after seeing a desired threshold. DISMISSED.** The polynomial and rational lower bound are unchanged from v1.0. Only the exact comparison $L > -4/5$ is newly extracted.
2. **Objection: the source 9I/10 was counted inside the owner. DISMISSED.** It remains outside (1.2). Rho is computed from $10/11 + \tilde{P}_{\text{lower}}$ exactly once.
3. **Objection: (8.5) refutes R-166. DISMISSED.** R-166 assumes one fresh $4p$ pair. The counter-direction uses simultaneous $4p$ and $8p$ support and refutes only coefficient-one tensorization of the lemma.
4. **Objection: failure of the carry-over polynomial proves the full two-root owner fails. UPHELD against that inference.** Equation (8.7) shows failure of the old adverse ledger with a weakened favorable constant, not the sign of every missing cross block.
5. **Objection: the finite hostile scan proves a negative eigenvalue. UPHELD against that inference.** It has convergence evidence but no outward error enclosure. It remains an unregistered internal falsifier.

6. **Objection:** this closes T-050 or selects a physical vacuum. **UPHELD** against that **inference**. The complete production incidence, root cross blocks, low variables, feedback, removal and physical comparison gates remain open. No morphology, phase or PDE follows.

11. Reproduction and Result footer

Result ID: A13-CLASSII-SPARSE-PRODUCTION-OWNER-RADIAL-GRAM-
GLOBAL-BOUNDARY (R-166 v1.1).
Precise statement: on the fixed exact p:2p past / fresh-4p fibre,
 $K_{\text{owner}}[g] \geq P_{\text{tilde}}(|g|) I$ and $K_{\text{owner}} > -4I/5$ for every g.
Scoped R-164 choice: $\rho=3/20$; full semiconvexity 31/220;
 $\epsilon_{\text{v}}=367/880$.
Exact method boundary: coefficient-one direct harmonic coercivity does not
tensorize to simultaneous fresh-4p and fresh-8p support; rational ratio 3/8.
Dependencies: A1; R-130, R-149, R-153, R-156, R-164, R-165.
Scope: fixed side-16 finite chart, positive floor, exact continuum torus
average, unit retained multipliers, declared p:2p past and one 4p root.
Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/a13_classii_sparse_production_owner_radial_gram_
global_boundary.py
Expected output: primary 40/40, independent 37/37, integrated verification,
deterministic PDF, and all release gates pass.
Independent command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/a13_classii_sparse_production_owner_radial_gram_
global_boundary_independent.py
Integrated command: E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/a13_classii_sparse_production_owner_radial_gram_
global_boundary_verify.py
Falsification gate: failure of the Gram split, resonance cancellation,
radial bounds, single-pair harmonic lemma, one-use ledger, rational minimum,
strengthened ρ arithmetic, non-tensorization witness, authorities, PDF,
records, or scope firewall.
Tier before / after: T4 / T4; no promotion.
Proof complete: false. T-050 closed: false. Sector A closed: false.
No-overclaim statement: no complete multi-root owner, feedback/removal, T-050, A13,
Nelson/measure, phase, morphology, PDE, or Sector A theorem.
Next required action: define one canonical finite production cylinder before computing
additional root and cross-root owners; continue truth-first Pre-A screening.